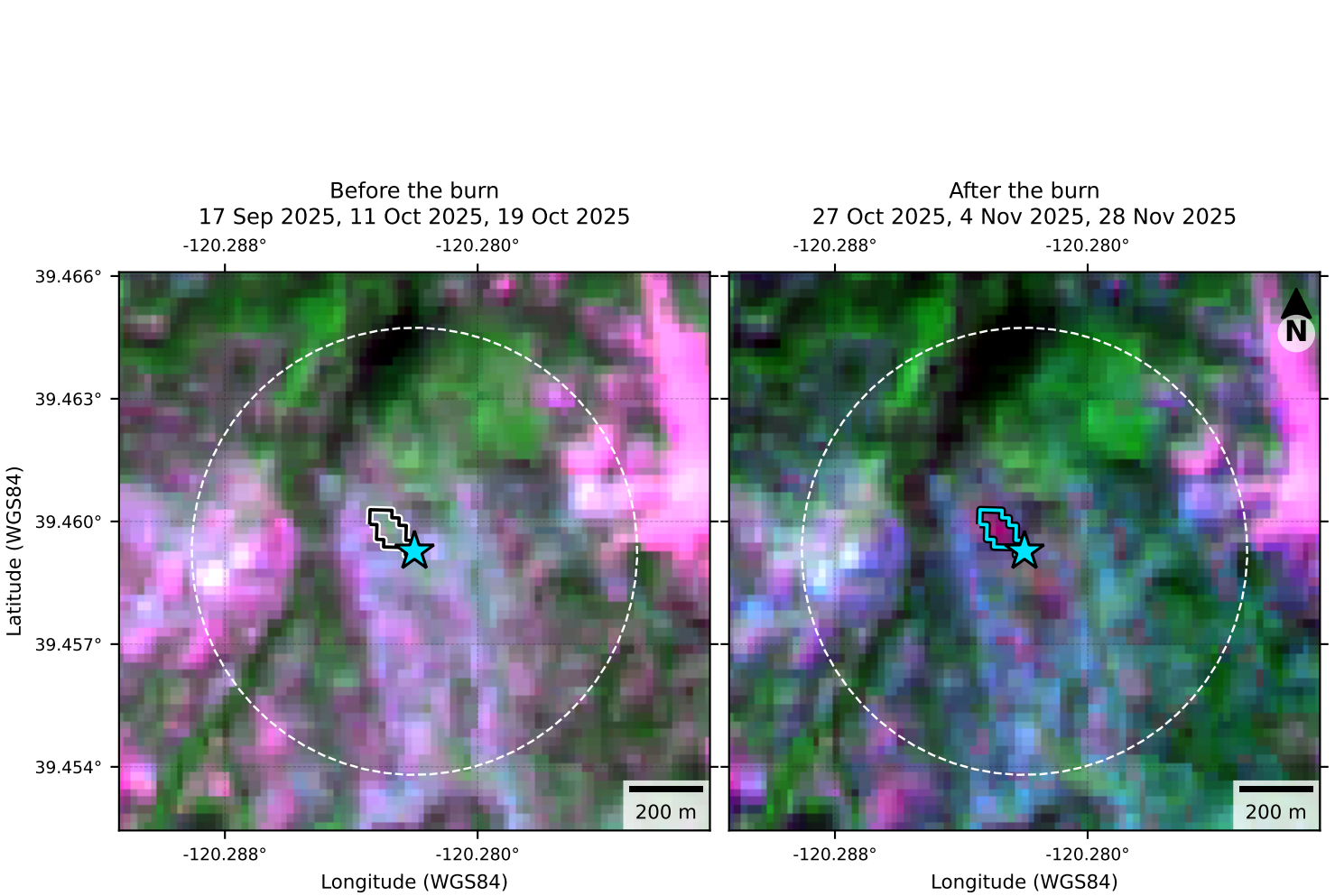
burn_3

Map 1 of 5: Landsat 8/9 before and after the burn | prescribed burn 21 Oct 2025 | reported ~32 ac (approx.)



Mapped: 2.0 ac (reported ~32 ac, approx.)

Images: 4 before, 3 after (searched 60 days before and 45 days after). Each panel is the per-pixel median of the 3 clearest looks closest to the burn; dates above are those looks. Flat grey = no clear view.

How the outline is drawn:
Landsat 8/9 (30 m) burn index drop that is next to the recorded point and dated to the burn.

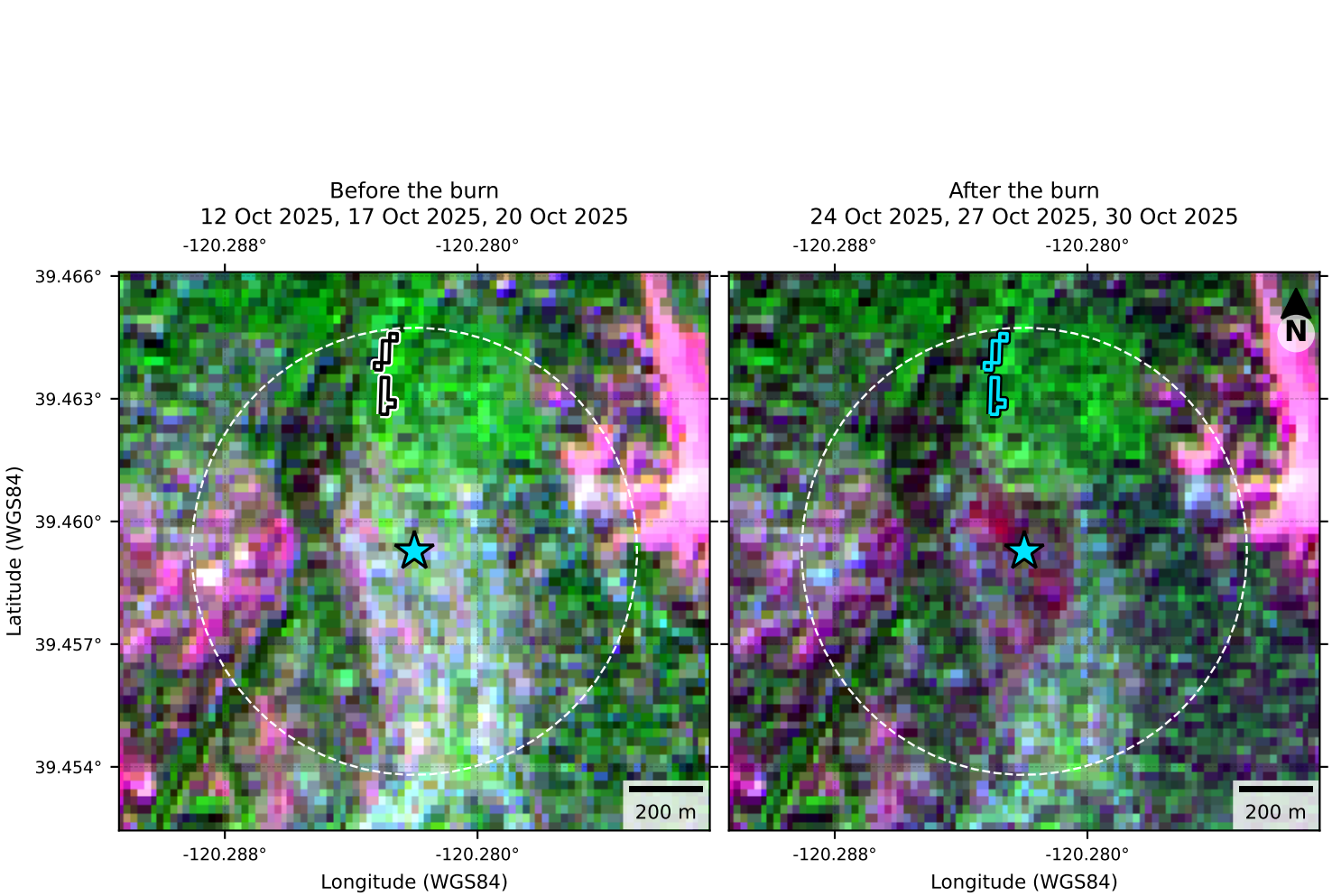
False colour (SWIR-NIR-red): healthy vegetation green, fresh burn scar dark red / brown / purple.



Satellite estimate from a recorded point, date and approximate acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Landsat 8/9 surface reflectance (30 m resampled to 20 m). Coordinates WGS84 decimal degrees.

burn_3

Map 2 of 5: Sentinel-2 before and after the burn | prescribed burn 21 Oct 2025 | reported ~32 ac (approx.)



Mapped: 1.1 ac (reported ~32 ac, approx.)

Images: 22 before, 16 after (searched 60 days before and 45 days after). Each panel is the per-pixel median of the 3 clearest looks closest to the burn; dates above are those looks. Flat grey = no clear view.

How the outline is drawn:
Sentinel-2 (20 m) burn index drop that is next to the recorded point and dated to the burn.

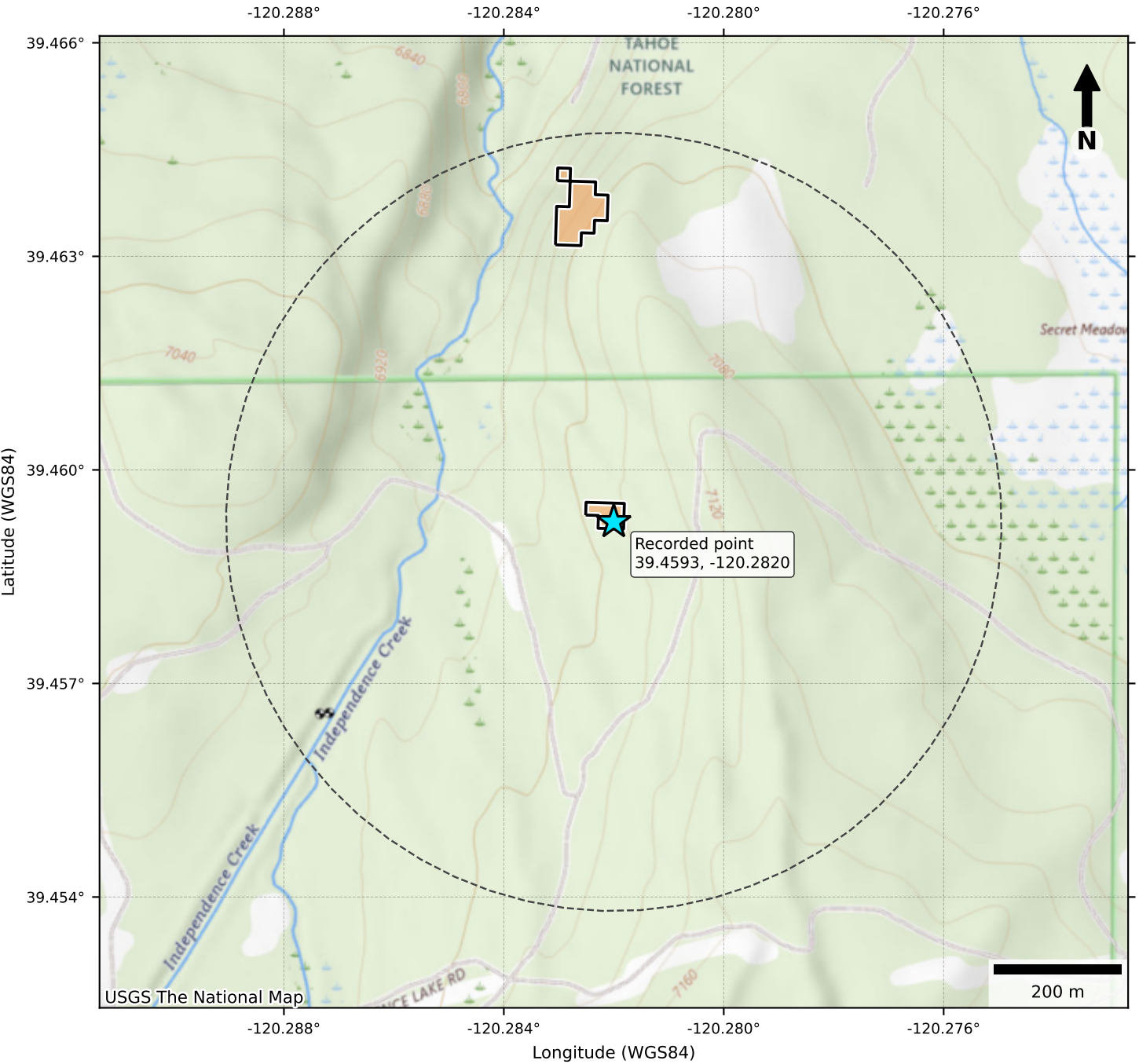
False colour (SWIR-NIR-red): healthy vegetation green, fresh burn scar dark red / brown / purple.



Satellite estimate from a recorded point, date and approximate acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Sentinel-2 surface reflectance (20 m). Coordinates WGS84 decimal degrees.

burn_3

Map 3 of 5: Sentinel-2 + Landsat combined | prescribed burn 21 Oct 2025 | reported ~32 ac (approx.)



Combined method

- Mapped burned area
- 1 km search radius around point
- Recorded lat/long

Mapped: 2.0 ac (reported ~32 ac, approx.)
Images: 22 + 16 Sentinel-2, 4 + 3 Landsat (before + afte
First clear image after burn: 3 days

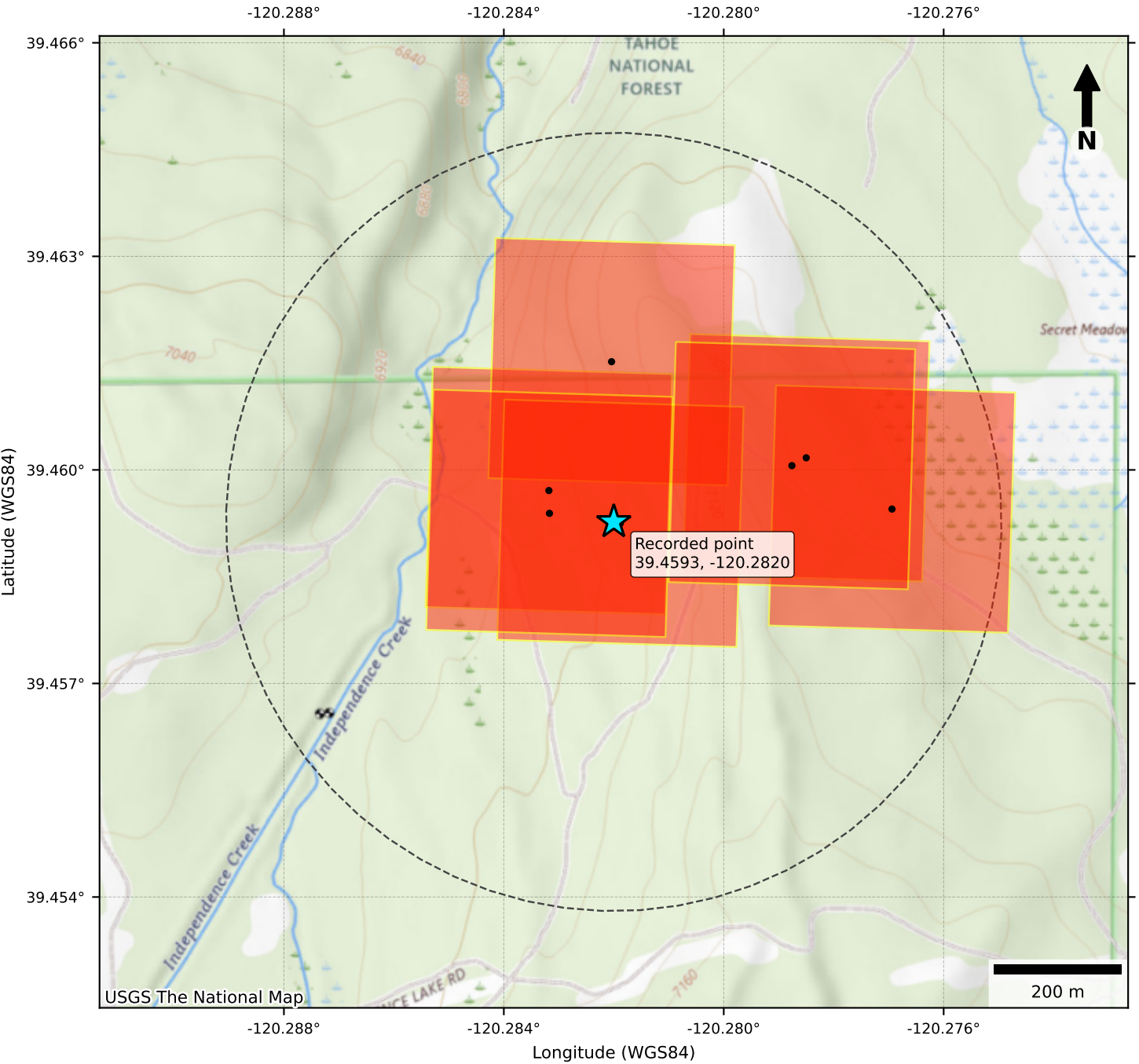
How the outline is drawn:
Both satellites in one time series (more clear looks).
Each image is compared with land far from the point,
so seasonal change cancels out. No embeddings.



Satellite estimate from a recorded point, date and approximate acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Combined Sentinel-2 + Landsat burn index, point + date match. Coordinates WGS84 decimal degrees.

burn_3

Map 4 of 5: VIIRS active-fire detections (NASA FIRMS) | prescribed burn 21 Oct 2025 | reported ~32 ac (approx.)



VIIRS active fire

- VIIRS fire detection (375 m pixel footprint)
- 1 km search radius around point
- Recorded lat/long

Window: 20 Oct - 23 Oct 2025
Detections within 1 km of the point (+ half a pixel): 7
All detections loaded: 11 detections within ~30 km
Dates: 21 Oct, 23 Oct

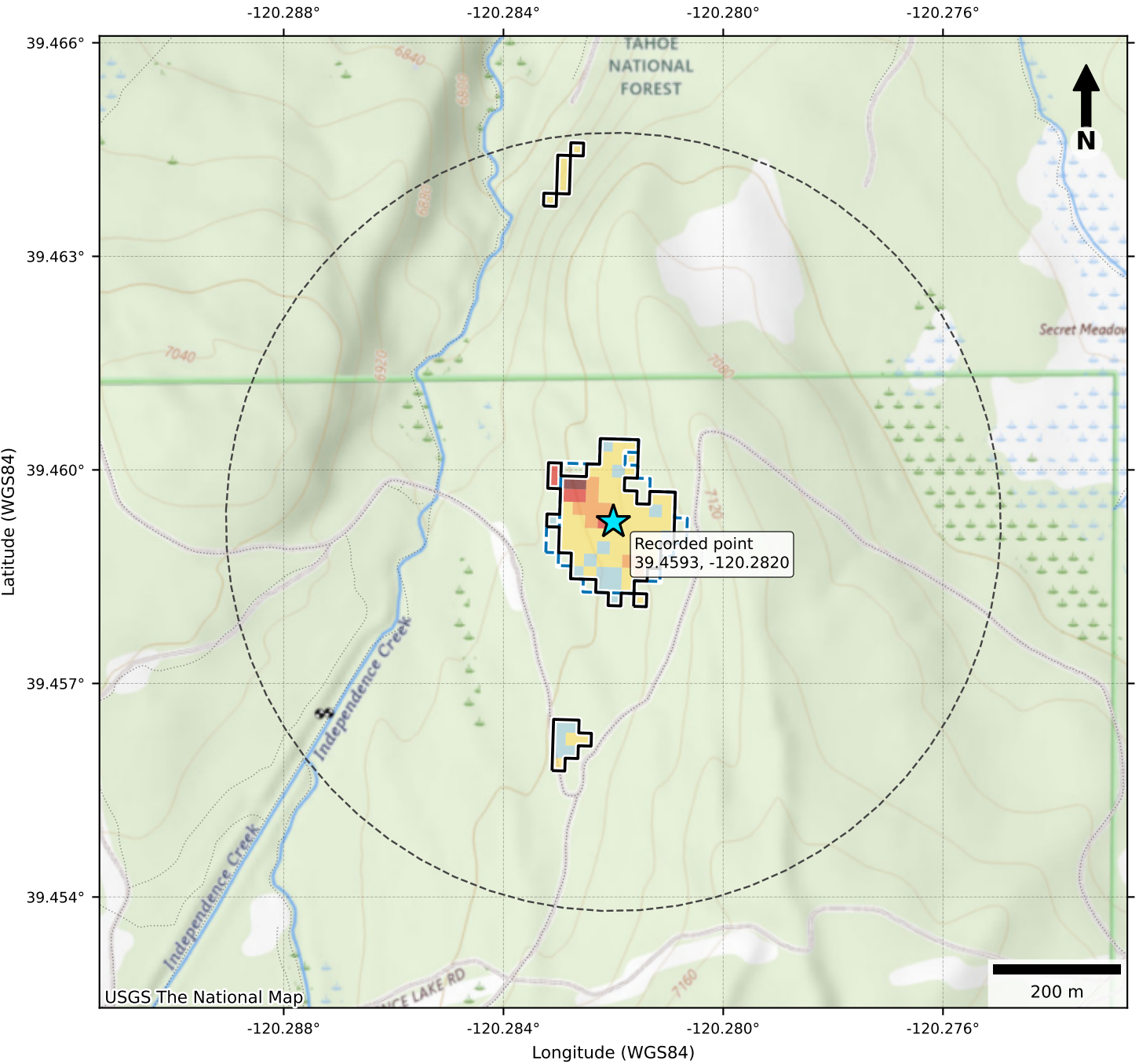
VIIRS (Suomi-NPP / NOAA-20 / NOAA-21) sees heat from active flames when a satellite passes (about twice a day per satellite). It confirms fire was present, not how far it spread; small, cool or short burns are often missed.



Satellite estimate from a recorded point, date and approximate acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
VIIRS 375 m active-fire product, NASA FIRMS. Coordinates WGS84 decimal degrees.

burn_3

Map 5 of 5: best effort (adds embeddings) | prescribed burn 21 Oct 2025 | reported ~32 ac (approx.)



Burn severity (satellite index)

- Unburned / very low (dNBR < 0.10): 2 ac
- Low (0.10 - 0.27): 6 ac
- Moderate-low (0.27 - 0.44): 1 ac
- Moderate-high (0.44 - 0.66): 0 ac
- High (> 0.66): 0 ac
- Mapped burn perimeter (best effort)
- Snapped to roads/trails/ridges (8.5 ac)
- Roads / trails / streams (Overture)
- 1 km search radius around point
- Recorded lat/long

Mapped: 9.1 ac (reported ~32 ac, approx.)
Canopy cover (USFS/FIA TCC): 39% (woodland / savanna)
Images: 38 Sentinel-2 + 7 Landsat
First clear image after burn: 3 days
Confidence: MEDIUM

How the outline is drawn:
As the combined method, but each pixel is compared only with look-alike land (same vegetation type, from TESSERA satellite embeddings of the year before), with delayed-scorch, SWIR char (NBR2), and seed-and-grow detection.



Satellite estimate from a recorded point, date and approximate acres (free data); not field-verified. Prescribed burns under canopy often read as unburned even where surface fuels burned.
Severity = green-up-corrected dNBR, Key & Benson (2006) classes, not calibrated to field (CBI) plots. Coordinates WGS84 decimal degrees.